WORKED SOLUTION

STEP II 2022, Question 6

Pure • Orthogonal Trajectories

STEP II 2022, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Two curves meet at right angles exactly when the product of their gradients there is -1. So for a whole family of curves we want a gradient formula that holds at a point (x, y) *whichever* member of the family passes through it. That means eliminating the parameter (a, b, c or k) between the curve's equation and its derivative, leaving a differential equation in x, y and $\frac{\mathrm{d}y}{\mathrm{d}x}$ alone. Then the perpendicular family is found by replacing the gradient m by $-1/m$ and solving.

Solution

Throughout, $x > 0$ and $y > 0$.

Part (i)(a)

Differentiating $x^2 + y^2 = 2ax$ implicitly with respect to x gives

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2a.$$

This still contains a, which changes from curve to curve. The curve's equation gives $2a = (x^2 + y^2)/x$, so substituting and multiplying by x,

$$2x^2 + 2xy\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2 \quad \Longrightarrow \quad 2xy\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 - x^2,$$

as required.

For the second family, differentiating $x^2 + y^2 = 2by$ gives (remembering that the right-hand side also depends on x through y)

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2b\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Now $2b = (x^2 + y^2)/y$; substituting and multiplying by y,

$$2xy + 2y^2\frac{\mathrm{d}y}{\mathrm{d}x} = (x^2 + y^2)\frac{\mathrm{d}y}{\mathrm{d}x} \quad \Longrightarrow \quad (x^2 - y^2)\frac{\mathrm{d}y}{\mathrm{d}x} = 2xy.$$

Part (i)(b)

Suppose a curve of each family passes through (x, y), with $y \neq x$. Since $x, y > 0$ we have $2xy \neq 0$ and, because $y \neq x$ and both are positive, $x^2 - y^2 \neq 0$. So we may divide in both differential equations: the two gradients are

$$m_1 = \frac{y^2 - x^2}{2xy}, \qquad m_2 = \frac{2xy}{x^2 - y^2}, \qquad \text{so} \qquad m_1 m_2 = \frac{y^2 - x^2}{x^2 - y^2} = -1.$$

Both gradients are finite and their product is -1, so the tangents are perpendicular.

Ask yourself What goes wrong on the line $y = x$, and does that mean the result fails there?

On $y = x$ the second equation reads $0 \cdot \frac{\mathrm{d}y}{\mathrm{d}x} = 2x^2 \neq 0$, which no finite gradient satisfies: the tangent to the second curve is vertical. That breaks the algebra, not the geometry, so we look directly.

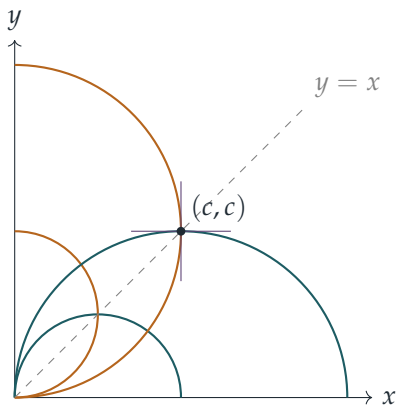
If (c, c) lies on $x^2 + y^2 = 2ax$ then $2c^2 = 2ac$, so $a = c$; similarly $b = c$. The first curve is the circle with centre $(c, 0)$ and radius c, and (c, c) is its highest point, so the tangent there is horizontal (indeed $m_1 = 0$ when $y = x$):

$$y = c.$$

The second curve is the circle with centre $(0, c)$ and radius c, and (c, c) is its right-most point, so its tangent is vertical:

$$x = c.$$

A horizontal and a vertical line are perpendicular, so **yes**: the tangents are perpendicular at every meeting point, including those on $y = x$.



Circles of the first family (teal) touch the y-axis at the origin; those of the second (amber) touch the x-axis there.

Part (ii)

For $y = c\ln x$ we have $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{c}{x}$. To eliminate c, note that at a point of the first quadrant on such a curve, $y > 0$ forces $\ln x \neq 0$, so $c = y/\ln x$ and

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x\ln x}.$$

This gradient is never zero, so a perpendicular curve through the same point must have gradient $-x\ln x/y$. The second family therefore satisfies

$$y\frac{\mathrm{d}y}{\mathrm{d}x} = -x\ln x.$$

Separating variables, and integrating $x\ln x$ by parts with $u = \ln x$, $\mathrm{d}v = x\,\mathrm{d}x$,

$$\int y\,\mathrm{d}y = -\int x\ln x\,\mathrm{d}x \quad\Longrightarrow\quad \tfrac{1}{2}y^2 = -\tfrac{1}{2}x^2\ln x + \tfrac{1}{4}x^2 + C.$$

Multiplying by 4, the second family is

$$2y^2 + 2x^2\ln x - x^2 = k,$$

one curve for each constant k (only the parts with $x > 0$, $y > 0$ count). Conversely, differentiating this implicitly gives $4y\frac{\mathrm{d}y}{\mathrm{d}x} + 4x\ln x + 2x - 2x = 0$, which is exactly the equation above, so wherever one of these curves meets some $y = c\ln x$ the product of the gradients is $\frac{y}{x\ln x}\cdot\frac{-x\ln x}{y} = -1$.

Part (iii)

Differentiating $y^2 = 4k(x+k)$ gives $2y\frac{\mathrm{d}y}{\mathrm{d}x} = 4k$, so, as $y > 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2k}{y}.$$

Ask yourself Here there is only one family. Where does the "second curve" come from?

Fix a point (x, y) in the first quadrant and ask which members of the family pass through it. The value of k must satisfy

$$4k^2 + 4xk - y^2 = 0,$$

a quadratic in k. Its discriminant $16x^2 + 16y^2$ is positive, so there are two distinct real roots k_1, k_2, and their product is

$$k_1k_2 = -\frac{y^2}{4}.$$

(This is negative, so both roots are non-zero, as the family requires.) A quadratic has at most two roots, so if two different curves of the family meet at (x, y), their parameters must be exactly k_1 and k_2. Their gradients there are $2k_1/y$ and $2k_2/y$, with product

$$\frac{4k_1k_2}{y^2} = \frac{4}{y^2} \cdot \left(-\frac{y^2}{4}\right) = -1.$$

So the tangents are perpendicular at every such point, as required. (These curves are parabolas with focus at the origin; the family is its own set of orthogonal trajectories.)

Checks

- In (i), take $(x, y) = (1, 2)$: then $a = \frac{5}{2}$, $b = \frac{5}{4}$, and the gradients are $m_1 = \frac{3}{4}$ and $m_2 = -\frac{4}{3}$, with product -1.
- In (ii), at $(e, 1)$ the log curve has $c = 1$ and gradient $1/e$; the formula for the second family gives gradient $-e \cdot 1/1 = -e$. Again the product is -1.
- In (iii), at $(0, 2)$ the quadratic is $4k^2 - 4 = 0$, so $k = \pm 1$: the parabolas $y^2 = 4(x + 1)$ and $y^2 = -4(x - 1)$, with gradients 1 and -1 there.
- All four results were also confirmed by computer at a few thousand random points, using numerical derivatives of the curves themselves.

Common slips

- Forgetting the factor $\frac{\mathrm{d}y}{\mathrm{d}x}$ on the right when differentiating $2by$.
- In (ii), writing the perpendicular gradient as $-x/c$ and integrating with c still present: c belongs to the first curve, so it must be eliminated before solving.
- Dividing by $x^2 - y^2$ in (i)(b) without saying why it is non-zero, and so missing that the line $y = x$ needs separate treatment.

Beyond STEP

Families of curves meeting everywhere at right angles are called *orthogonal trajectories*. They appear in physics as field lines and equipotentials, and in complex analysis: the real and imaginary parts of an analytic function always have perpendicular level curves, which is why the circles in (i) arise from $1/z$ and the self-orthogonal parabolas in (iii) from $\sqrt{z}$. Boyce and DiPrima, *Elementary Differential Equations*, treats orthogonal trajectories, and Needham, *Visual Complex Analysis*, explains the complex-analysis picture beautifully.

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